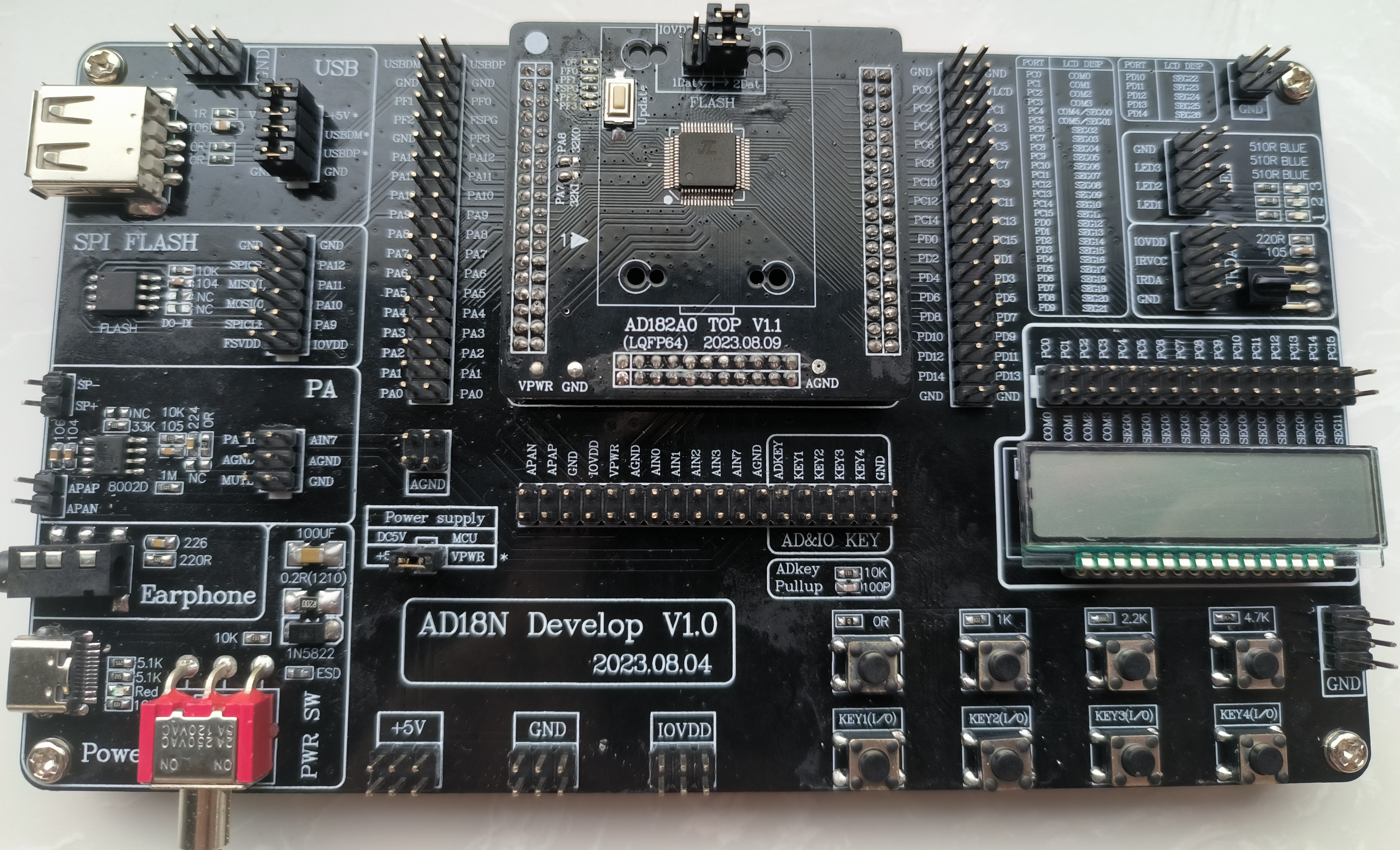
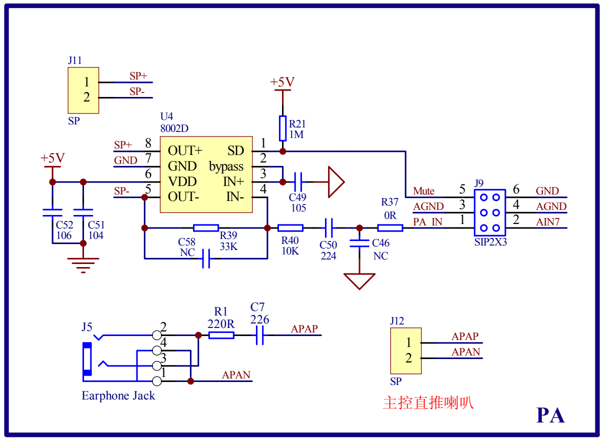
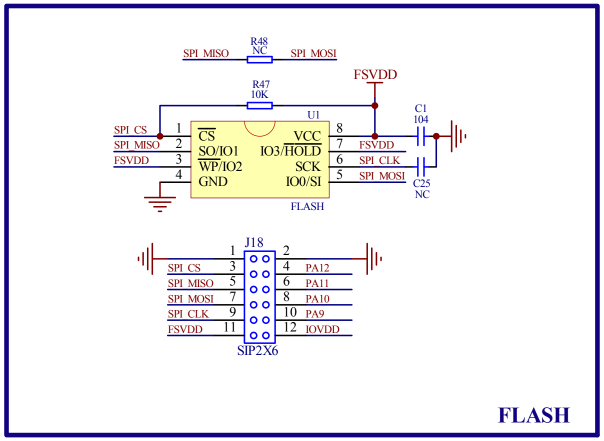
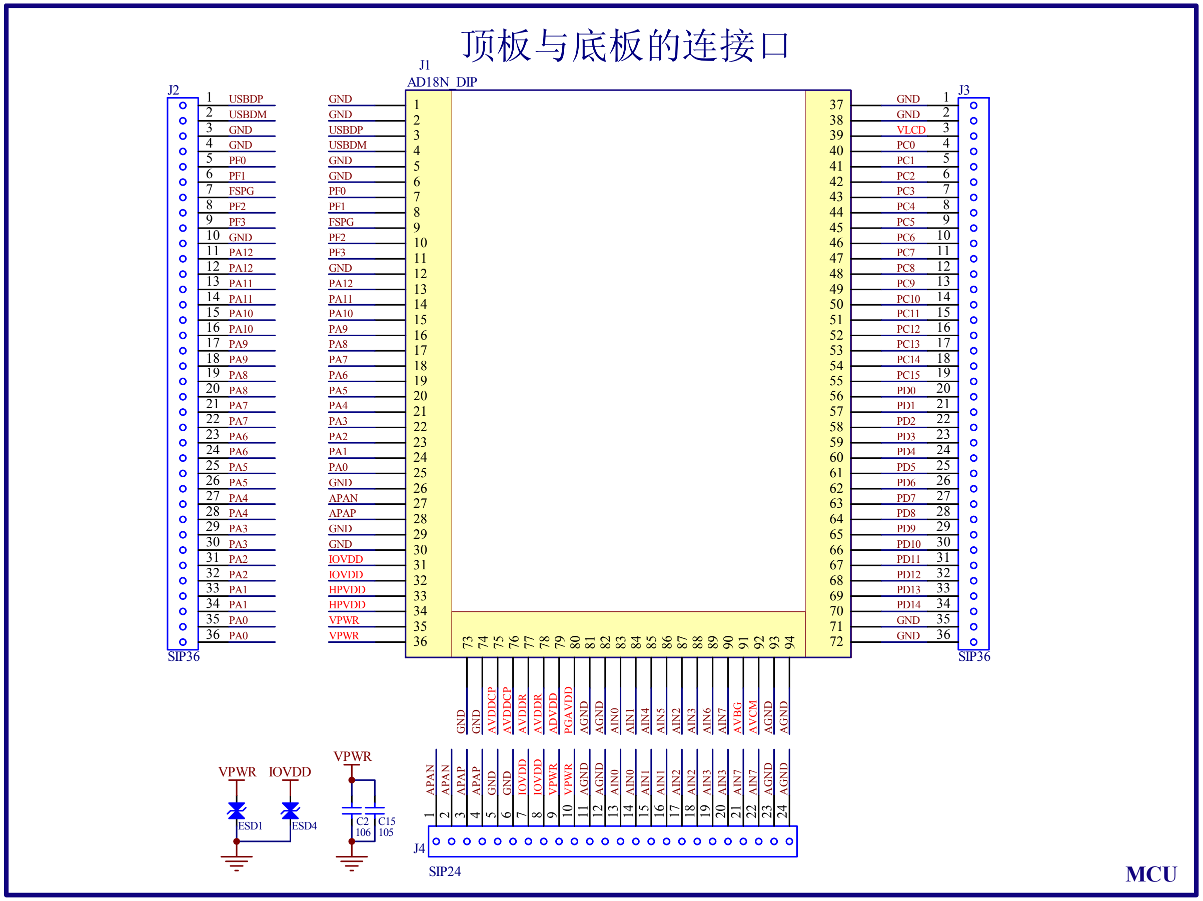
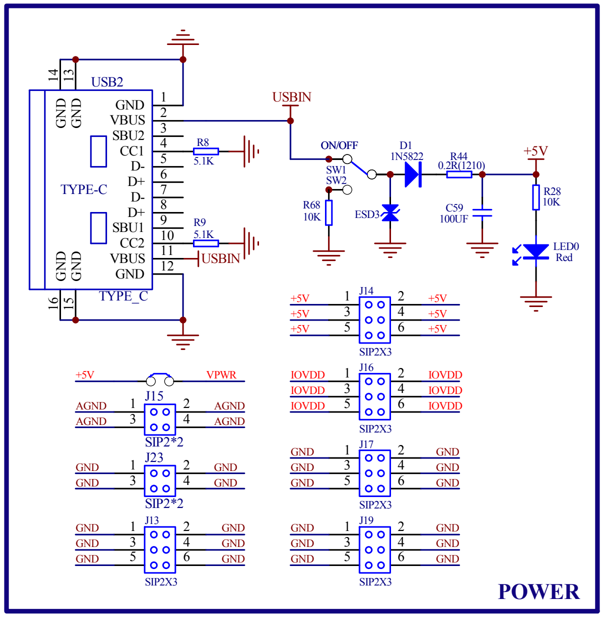

AD18N Develop V1.0
2023.08.04
AD182A0 TOP V1.1
(LQFP64) 2023.08.09
USB
GND
+5V
USBDM
USBDP
1R
106
0R
SPI FLASH
FLASH
SPICS
MISO
MOSI
SPICLK
FSVDD
10K
104
NC
DO-DI
PA
SP-
SP+
PA_in
AGND
MUTE
APAP
APAN
8002D
33K
105
224
1M
AIN7
Earphone
226
220R
100UF
0.2R(1210)
1N5822
ESD
10K
5.1K
Red
Power
PWR SW
Power supply
DC5V
MCU
VPWR
+5V
IOVDD
USBDM
USBDP
PF0
PF1
PF2
PF3
FSPG
PA12
PA11
PA10
PA9
PA8
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
FLASH
1Dat
2Dat
Update
32K0
32K1
VPWR
AGND
GND
PC0
PC2
PC4
PC6
PC8
PC10
PC12
PC14
PD0
PD2
PD4
PD6
PD8
PD10
PD12
PD14
VLCD
PC1
PC3
PC5
PC7
PC9
PC11
PC13
PC15
PD1
PD3
PD5
PD7
PD9
PD11
PD13
PORT
LCD DISP
COM0
COM1
COM2
COM3
COM4/SEG00
COM5/SEG01
SEG02
SEG03
SEG04
SEG05
SEG06
SEG07
SEG08
SEG09
SEG10
SEG11
SEG12
SEG13
SEG14
SEG15
SEG16
SEG17
SEG18
SEG19
SEG20
SEG21
PD10
PD11
PD12
PD13
PD14
SEG22
SEG23
SEG24
SEG25
SEG26
LED3
LED2
LED1
510R BLUE
IOVDD
IRVCC
IRDA
220R
105
APAN
APAP
AIN0
AIN1
AIN2
AIN3
ADKEY
KEY1
KEY2
KEY3
KEY4
AD&IO KEY
ADkey Pullup
10K
100P
0R
1K
2.2K
4.7K
KEY1(I/O)
KEY2(I/O)
KEY3(I/O)
KEY4(I/O)

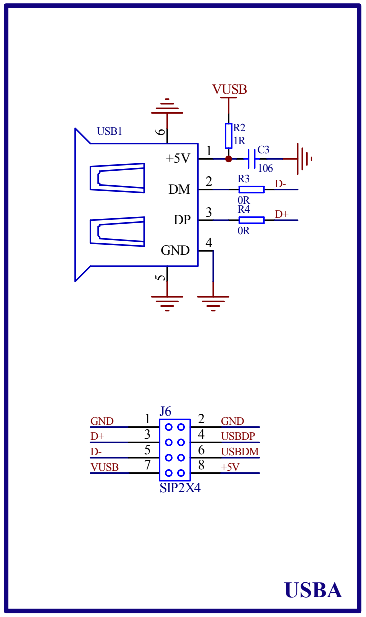

J11
SP+
SP-
SP
U4
8002D
OUT+
SD
GND
bypass
VDD
IN+
OUT-
IN-
+5V
R21
1M
C49
105
R37
0R
Mute
AGND
PA_IN
J9
SIP2X3
GND
AGND
AIN7
C52
106
C51
104
C58
NC
R39
33K
R40
10K
C50
224
C46
NC
J5
R1
220R
C7
226
APAP
APAN
Earphone Jack
J12
SP
主控直推喇叭
PA

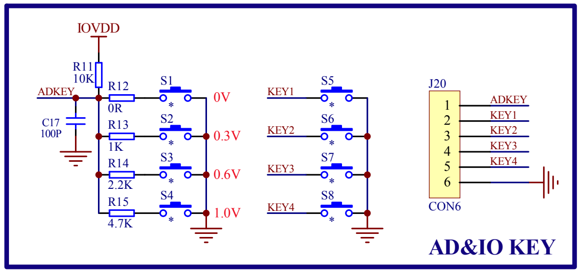

SPI_MISO
R48
NC
SPI_MOSI
R47
10K
FSVDD
U1
C1
104
SPI_CS
CS
VCC
SO/IO1
IO3/HOLD
WP/IO2
SCK
GND
IO0/SI
SPI_CLK
C25
NC
FLASH
J18
PA12
PA11
PA10
PA9
IOVDD
SIP2X6
FLASH

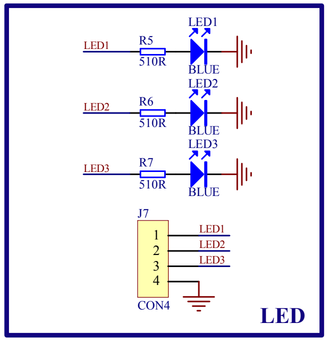
顶板与底板的连接口

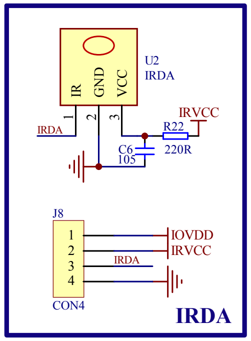

J1
AD18N_DIP
J2
SIP36
J3
SIP36
USBDP
USBDM
GND
PF0
PF1
FSPG
PF2
PF3
PA12
PA11
PA10
PA9
PA8
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
APAN
APAP
IOVDD
HPVDD
VPWR
VLCD
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
AVDDCP
AVDDR
ADVDD
PGAVDD
AGND
AIN0
AIN1
AIN4
AIN5
AIN2
AIN3
AIN6
AIN7
AVBG
AVCM
VPWR
IOVDD
ESD1
ESD4
C2
106
C15
105
J4
SIP24
MCU

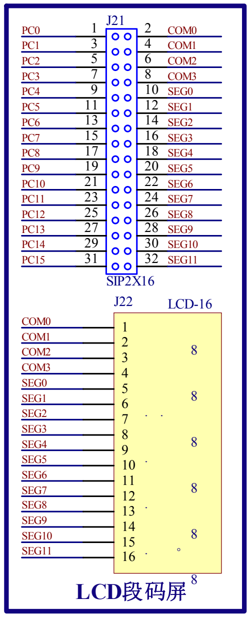

USB2
TYPE-C
GND
VBUS
SBU2
CC1
D-
D+
SBU1
CC2
TYPE_C
R8
5.1K
R9
5.1K
USBIN
ON/OFF
SW1
SW2
D1
1N5822
R44
0.2R(1210)
+5V
R68
10K
ESD3
C59
100UF
R28
10K
LED0
Red
J14
SIP2X3
J15
SIP2*2
AGND
VPWR
J16
IOVDD
J17
J23
J13
J19
POWER

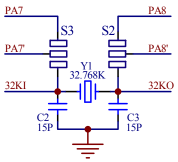

VUSB
USB1
+5V
DM
DP
GND
R2
1R
C3
106
R3
0R
R4
0R
D-
D+
J6
SIP2X4
GND
USBDP
USBDM
VUSB
USBA

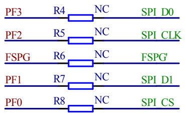

IOVDD
R11
10K
ADKEY
C17
100P
R12
0R
R13
1K
R14
2.2K
R15
4.7K
S1
S2
S3
S4
0V
0.3V
0.6V
1.0V
KEY1
KEY2
KEY3
KEY4
S5
S6
S7
S8
J20
CON6
AD&IO KEY

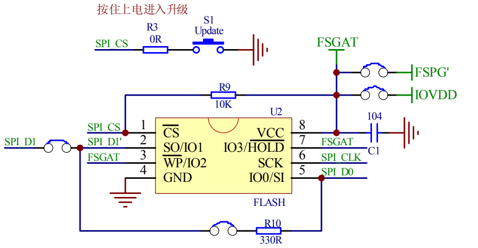

LED1
LED2
LED3
R5
510R
R6
510R
R7
510R
BLUE
J7
CON4
LED


U2
IRDA
IR
GND
VCC
IRVCC
R22
220R
C6
105
J8
CON4
IOVDD
IRVCC
IRDA

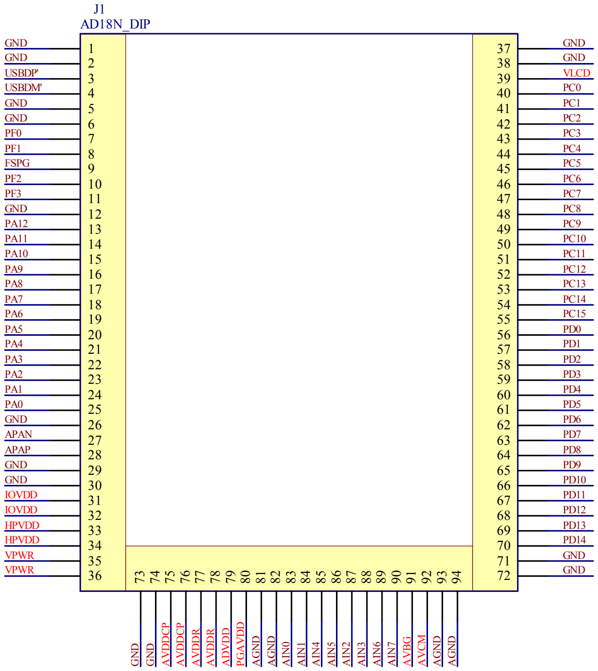

J21
SIP2X16
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
COM0
COM1
COM2
COM3
SEG0
SEG1
SEG2
SEG3
SEG4
SEG5
SEG6
SEG7
SEG8
SEG9
SEG10
SEG11
J22
LCD-16
LCD段码屏

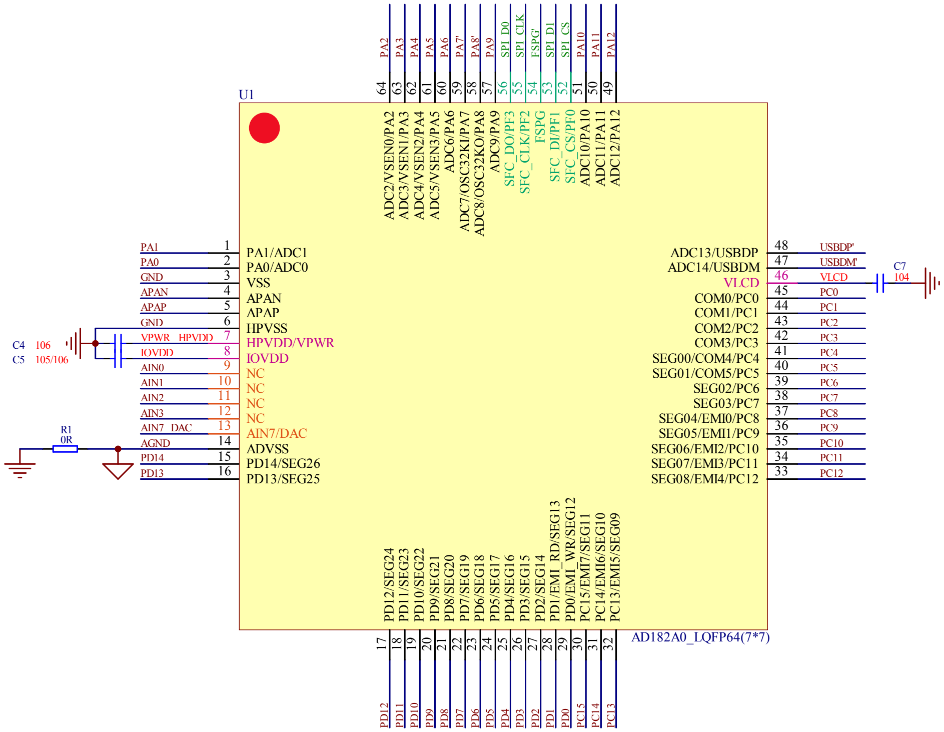

PA7
PA8
S3
S2
PA7'
PA8'
Y1
32.768K
32KI
32KO
C2
15P
C3
15P

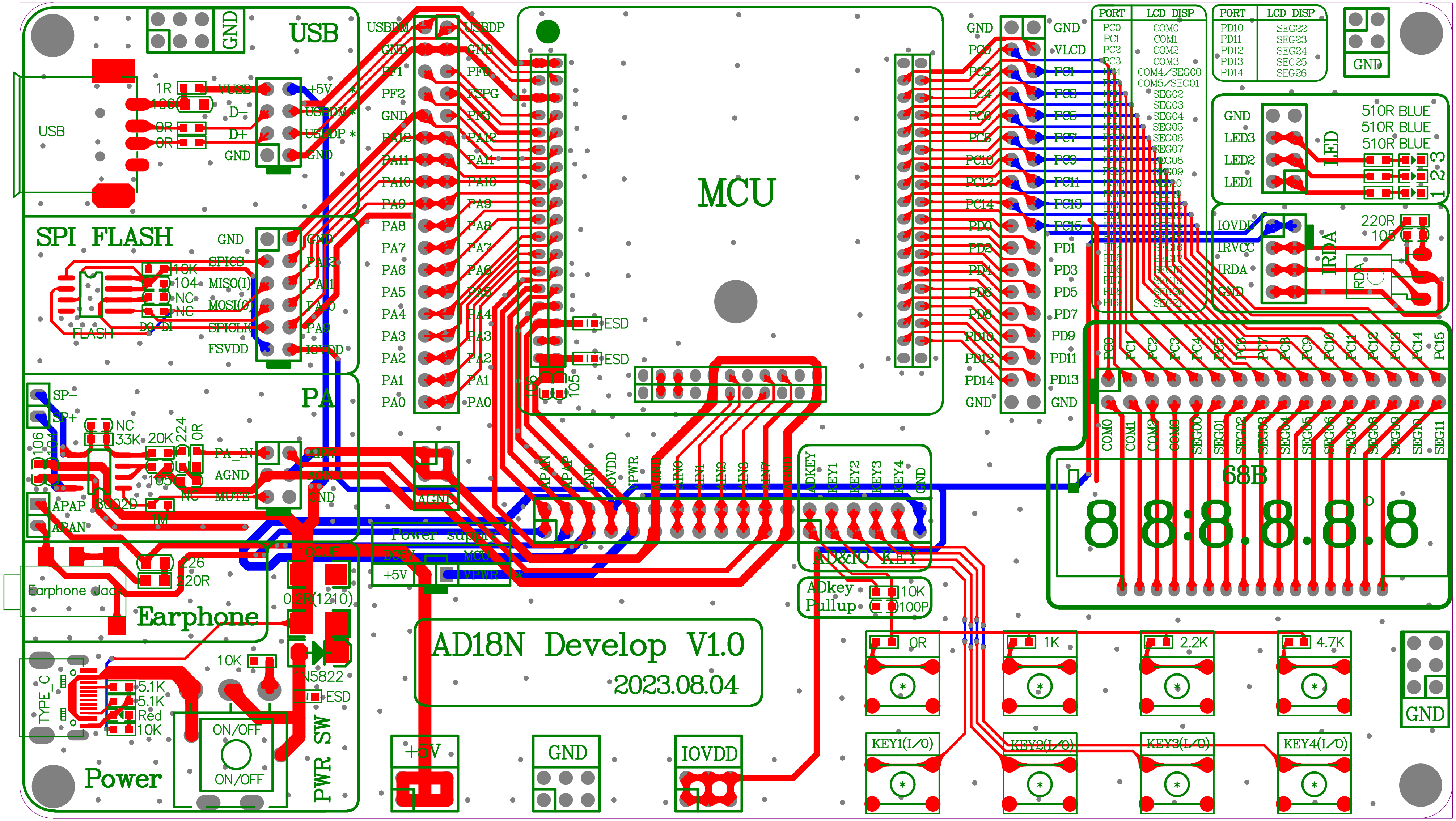

PF3
R4
NC
SPI_D0
PF2
R5
NC
SPI_CLK
FSPG
R6
NC
FSPG'
PF1
R7
NC
SPI_D1
PF0
R8
NC
SPI_CS

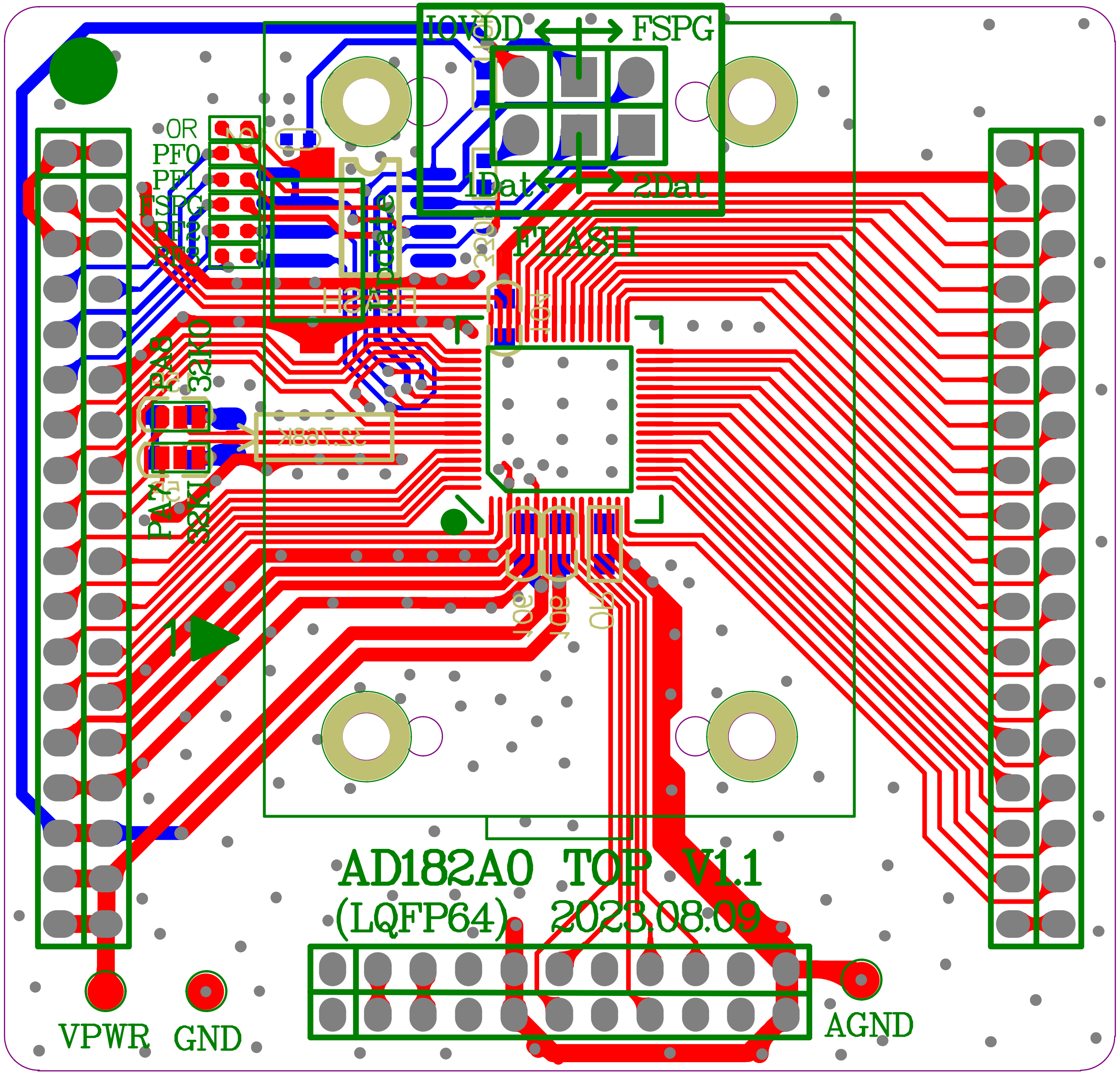

按住上电进入升级
R3
0R
S1
Update
SPI_CS
FSGAT
FSPG'
IOVDD
R9
10K
U2
104
C1
SPI_D1
SPI_CS
SPI_D1'
FSGAT
1
2
3
4
CS
SO/IO1
WP/IO2
GND
VCC
IO3/HOLD
SCK
IO0/SI
8
7
6
5
FSGAT
SPI_CLK
SPI_D0
FLASH
R10
330R

顶板与底板的连接口


J1
AD18N_DIP
GND 1
GND 2
USBDP' 3
USBDM' 4
GND 5
GND 6
PF0 7
PF1 8
FSPG 9
PF2 10
PF3 11
GND 12
PA12 13
PA11 14
PA10 15
PA9 16
PA8 17
PA7 18
PA6 19
PA5 20
PA4 21
PA3 22
PA2 23
PA1 24
PA0 25
GND 26
APAN 27
APAP 28
GND 29
GND 30
IOVDD 31
IOVDD 32
HPVDD 33
HPVDD 34
VPWR 35
VPWR 36
37 GND
38 GND
39 VLCD
40 PC0
41 PC1
42 PC2
43 PC3
44 PC4
45 PC5
46 PC6
47 PC7
48 PC8
49 PC9
50 PC10
51 PC11
52 PC12
53 PC13
54 PC14
55 PC15
56 PD0
57 PD1
58 PD2
59 PD3
60 PD4
61 PD5
62 PD6
63 PD7
64 PD8
65 PD9
66 PD10
67 PD11
68 PD12
69 PD13
70 PD14
71 GND
72 GND
73 GND
74 GND
75 AVDDCP
76 AVDDCP
77 AVDDR
78 AVDDR
79 ADVDD
80 PGAVDD
81 AGND
82 AGND
83 AIN0
84 AIN1
85 AIN4
86 AIN5
87 AIN2
88 AIN3
89 AIN6
90 AIN7
91 AVBG
92 AVCM
93 AGND
94 AGND


U1
PA2 64 ADC2/VSEN0/PA2
PA3 63 ADC3/VSEN1/PA3
PA4 62 ADC4/VSEN2/PA4
PA5 61 ADC5/VSEN3/PA5
PA6 60 ADC6/PA6
PA7' 59 ADC7/OSC32KI/PA7
PA8' 58 ADC8/OSC32KO/PA8
PA9 57 ADC9/PA9
SPI_D0 56 SFC_DO/PF3
SPI_CLK 55 SFC_CLK/PF2
FSPG' 54 FSPG
SPI_D1 53 SFC_DI/PF1
SPI_CS 52 SFC_CS/PF0
PA10 51 ADC10/PA10
PA11 50 ADC11/PA11
PA12 49 ADC12/PA12
PA1 1 PA1/ADC1
PA0 2 PA0/ADC0
GND 3 VSS
APAN 4 APAN
APAP 5 APAP
GND 6 HPVSS
VPWR HPVDD 7 HPVDD/VPWR
IOVDD 8 IOVDD
AIN0 9 NC
AIN1 10 NC
AIN2 11 NC
AIN3 12 NC
AIN7_DAC 13 AIN7/DAC
AGND 14 ADVSS
PD14 15 PD14/SEG26
PD13 16 PD13/SEG25
C4 106
C5 105/106
R1
0R
ADC13/USBDP 48 USBDP'
ADC14/USBDM 47 USBDM'
VLCD 46 VLCD
COM0/PC0 45 PC0
COM1/PC1 44 PC1
COM2/PC2 43 PC2
COM3/PC3 42 PC3
SEG00/COM4/PC4 41 PC4
SEG01/COM5/PC5 40 PC5
SEG02/PC6 39 PC6
SEG03/PC7 38 PC7
SEG04/EMI0/PC8 37 PC8
SEG05/EMI1/PC9 36 PC9
SEG06/EMI2/PC10 35 PC10
SEG07/EMI3/PC11 34 PC11
SEG08/EMI4/PC12 33 PC12
C7
104
PD12/SEG24 17 PD12
PD11/SEG23 18 PD11
PD10/SEG22 19 PD10
PD9/SEG21 20 PD9
PD8/SEG20 21 PD8
PD7/SEG19 22 PD7
PD6/SEG18 23 PD6
PD5/SEG17 24 PD5
PD4/SEG16 25 PD4
PD3/SEG15 26 PD3
PD2/SEG14 27 PD2
PD1/EMI_RD/SEG13 28 PD1
PD0/EMI_WR/SEG12 29 PD0
PC15/EMI7/SEG11 30 PC15
PC14/EMI6/SEG10 31 PC14
PC13/EMI5/SEG09 32 PC13
AD182A0_LQFP64(7*7)


USB
GND
1R
106
0R
0R
VUSB
D-
D+
GND
+5V
USBDM
USBDP
GND
SPI FLASH
GND
SPICS
10K
104
NC
NC
MISO(I)
MOSI(O)
SPICLK
FSVDD
FLASH
DO
DI
GND
PA12
PA11
PA10
PA9
IOVDD
PA
SP-
SP+
NC
33K
20K
224
0R
106
PA_IN
AGND
MUTE
105
NC
APAP
APAN
8002D
1M
AGND
Earphone Jack
226
220R
Earphone
100UF
0.2R(1210)
1N5822
ESD
10K
TYPE_C
5.1K
5.1K
Red
10K
ON/OFF
ON/OFF
PWR SW
Power
Power supply
D-5V
MCU
+5V
VPWR
USBDM
USBDP
GND
GND
PF1
PF0
PF2
FSPG
GND
PF3
PA12
PA12
PA11
PA11
PA10
PA10
PA9
PA9
PA8
PA8
PA7
PA7
PA6
PA6
PA5
PA5
PA4
PA4
PA3
PA3
PA2
PA2
PA1
PA1
PA0
PA0
MCU
ESD
ESD
106
105
APAN
APAP
GND
IOVDD
VPWR
AGND
LIN0
LIN1
LIN2
LIN3
LIN4
AGND
ADKEY
KEY1
KEY2
KEY3
KEY4
GND
AD&IO KEY
ADkey
Pullup
10K
100P
AD18N Develop V1.0
2023.08.04
+5V
GND
IOVDD
GND
GND
PC0
VLCD
PC2
PC1
PC4
PC3
PC6
PC5
PC8
PC7
PC10
PC9
PC12
PC11
PC14
PC13
PD0
PC15
PD2
PD1
PD4
PD3
PD6
PD5
PD8
PD7
PD10
PD9
PD12
PD11
PD14
PD13
GND
GND
PORT
LCD DISP
PC0
COM0
PC1
COM1
PC2
COM2
PC3
COM3
PC4
COM4/SEG00
PC5
COM5/SEG01
SEG02
SEG03
SEG04
SEG05
SEG06
SEG07
SEG08
SEG09
PORT
LCD DISP
PD10
SEG22
PD11
SEG23
PD12
SEG24
PD13
SEG25
PD14
SEG26
GND
GND
LED3
LED2
LED1
LED
510R BLUE
510R BLUE
510R BLUE
1 2 3
IOVDD
IRVCC
IRDA
GND
IRDA
220R
105
IRDA
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14
PC15
COM0
COM1
COM2
COM3
SEG00
SEG01
SEG02
SEG03
SEG04
SEG05
SEG06
SEG07
SEG08
SEG09
SEG10
SEG11
68B
8 8:8.8.8.8
0R
1K
2.2K
4.7K
KEY1(I/O)
KEY2(I/O)
KEY3(I/O)
KEY4(I/O)
GND


IOVDD
FSPG
1Dat
2Dat
FLASH
0R
PF0
PF1
FSPG
PF2
PF3
Update
PA8
32K0
PA7
32K1
32.768K
15P
15P
10K
330k
104
104
1
AD182A0 TOP V1.1
(LQFP64) 2023.08.09
VPWR
GND
AGND